

ELEC5305 Project Proposal

The Effect of Speech Enhancement on Whisper Speech Recognition

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1. Project Overview

This project studies whether reducing noise before speech recognition can improve recognition accuracy. It focuses on Whisper and asks whether speech enhancement helps it recognise words correctly, or whether the enhancement process changes the speech too much and makes recognition worse.

Clean speech with known transcripts will be mixed with noise at different levels. Each noisy recording will be tested using four approaches: no enhancement, spectral subtraction, Wiener filtering and a pretrained DeepFilterNet model. The same Whisper model and settings will be used for all recordings to ensure a fair comparison. Word Error Rate (WER) will be the main measure of recognition accuracy. Additional analysis of speech quality and audio features will help explain why enhancement improves or reduces accuracy under different noise conditions.

2. Background and Motivation

Speech enhancement reduces background noise, but it may also remove parts of the speech. Spectral subtraction can create unwanted sounds known as “musical noise”. Wiener filtering relies on estimates of speech and noise power. Whisper is trained on a wide range of audio and can recognise speech in noisy conditions. This means that clearer audio does not always lead to more accurate recognition.

This project explores a practical question: when a microphone records noisy speech, does adding a denoising step help the recogniser? Denoising may improve audio quality while making quiet consonants or word endings harder to recognise. A pretrained DeepFilterNet model will be compared with the two classical methods, without training a new model.

The project will apply signal-processing techniques covered in the course, including STFT analysis, noise estimation, gain calculation and signal reconstruction. It will also examine how enhancement changes the speech. The aim is to provide a fair comparison that others can repeat and identify when enhancement helps or harms recognition.

3. Proposed Methodology

MATLAB will be used to prepare the audio, implement spectral subtraction and Wiener filtering, reconstruct the signals and evaluate audio quality. Python will be used to run DeepFilterNet and Whisper, calculate Word Error Rate (WER) and

analyse Whisper’s audio features. The first experiment will use ten Mini LibriSpeech recordings with reference transcripts, mixed with office noise at 0 dB SNR.

Recognition results will be compared for clean speech, noisy speech and speech processed by each of the three enhancement methods.

The main experiment will use 50–100 recordings from several speakers. A separate small development set will be used to choose the processing parameters. Three noise types will be tested: white noise generated in MATLAB, office noise from DEMAND and another DEMAND noise, such as traffic [3,4]. Each recording will be tested at 0, 5 and 10 dB SNR. With three noise types and four processing approaches, this gives 36 conditions per recording. If recognition is almost perfect at these noise levels, –5 dB may also be tested. The audio files, noise channels, noise starting positions and random seeds will be recorded so that the experiment can be repeated.

Audio will be processed in mono at 16 kHz. Noise will be added at the required SNR, and the volume will be adjusted to avoid clipping. A short noise-only segment will be added at the beginning to estimate noise power. Spectral subtraction and Wiener filtering will use the same noisy recordings, Hann window, STFT settings and initial noise estimates. Spectral subtraction will include a minimum power level to prevent excessive removal. Wiener filtering will calculate its gains from the estimated speech and noise power [5]. Both methods will keep the original noisy phase and reconstruct the audio using the inverse STFT and overlap-add. Clean speech will only be used to create the mixtures and evaluate the results.

DeepFilterNet will use a pretrained model without further training. Audio will be resampled from 16 to 48 kHz before enhancement and returned to 16 kHz afterwards. The model version, resampling method and handling of processing delay will be recorded. The same Whisper model, initially base.en, will be used throughout, with the same language, decoding and text-normalisation settings. Neither DeepFilterNet nor Whisper will be trained. Processing parameters will be chosen using the development set and kept unchanged during the final evaluation.

4. Expected Outcomes

WER will be calculated using JiWER and reference transcripts, with substitution, deletion and insertion errors reported separately. Overall WER will use the total word errors divided by the total reference words. Clean-speech WER will serve as a baseline. Enhancement will be evaluated using $\Delta\text{WER} = \text{enhanced WER} - \text{noisy WER}$, where negative values indicate improvement.

Output SNR, SNR improvement and STOI will help explain the results. Signals will be aligned, and the initial noise-only section will be excluded. Whisper's official preprocessing will be used to compare processed and clean log-Mel features using mean squared error. Smaller feature differences do not necessarily mean better recognition.

Figures will show the relationships between WER, input SNR, STOI and log-Mel feature differences. Selected cases where enhancement reduces accuracy will be analysed using spectrograms, transcripts and informal listening. Processing time and real-time factor will also be reported. Conclusions will be limited to the tested conditions, and informal listening will not be treated as a formal listening test.

5. Timeline (Weeks 1–13)

Week	Task
1–2	Consider Project Topic
3–5	Literature review and dataset collection
6–9	Initial implementation and testing
10–11	Optimisation and evaluation
12–13	Final report and GitHub documentation

6. References

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